VTPEH 6270 Checkpoint 3

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1 INTRODUCTION

This R Markdown document presents an exploratory analysis of the edible_plants Data, which contains structured observational data on edible plant species. This data set was uploaded under rfordatascience/tidytuesday on 2026-02-03. The dataset includes taxonomic information, cultivation characteristics, environmental requirements, and basic nutritional properties. Variables describe biologically relevant factors such as sunlight and water needs, soil preferences, temperature tolerance, germination duration, harvest time, and energy content.

The analysis is designed to examine scientifically plausible associations between cultivation conditions and plant growth outcomes. Emphasis is placed on reproducible data preparation, grouped summary statistics, and clear visualization to support interpretable and publication-quality reporting.

2 RESEARCH QUESTION AND BACKGROUND

2.1 RESEARCH QUESTION

Do plant water requirements differ across temperature tolerance classifications?

2.2 SCIENTIFIC PLAUSABILITY AND CONTEXT

Temperature is a major determinant of plant water balance because it regulates transpiration, vapor pressure gradients, and metabolic activity [@taiz2015plant]. Species adapted to warmer conditions often experience greater evaporative losses, which may necessitate higher water requirements, whereas cold-tolerant plants exhibit mechanisms that conserve water under reduced thermal stress [@taiz2015plant; @lambers2008plant]. These physiological relationships provide a biological basis for expecting differences in water needs across temperature tolerance classifications [@nobel2009physicochemical].

3 DATA PREPARATION

```
data_edible_plants <- read.csv("edible_plants (1).csv")
```

Data preparation focused on ensuring that the variables relevant to the research question were correctly structured for categorical analysis, enabling grouped summaries and visualization. The variables water and temperature_class were verified as categorical factors. Records with missing values were removed to prevent distortion of comparisons. No additional transformations or derived variables were required because both variables are already meaningful categories suitable for analysis. Continuous variables, if analyzed later, could be discretized as needed.

3.1 CLEANING

```
library(dplyr)
library(stringr)

data_edible_plants_clean <- data_edible_plants %>%
  mutate(
    water = tolower(trimws(water)),
    temperature_class = tolower(trimws(temperature_class))
  ) %>%
  filter(
    !is.na(water),
    !is.na(temperature_class)
  ) %>%
  mutate(
    water = factor(str_to_title(water)),
    temperature_class = factor(str_to_title(temperature_class))
  )
```

4 GROUPED SUMMARY STATISTICS

4.1 WATER REQUIREMENTS BY TEMPERATURE CLASS

```
library(dplyr)
library(knitr)
library(stringr)
```

```r
# Clean category names safely
data_clean <- data_edible_plants_clean %>%
  rename(
    Water = water,
    Temperature_Class = temperature_class
  ) %>%
  mutate(
    Water = str_to_title(Water),
    Temperature_Class = str_to_title(Temperature_Class)
  )

# Count species per combination and compute proportion within each temperature class
water_temp_prop <- data_clean %>%
  group_by(Temperature_Class, Water) %>%
  summarise(Count = n(), .groups = "drop") %>%
  group_by(Temperature_Class) %>%
  mutate(Proportion = round(Count / sum(Count), 2)) %>%
  filter(Count > 0) %>%
  arrange(Temperature_Class, Water)

# Display table
kable(
  water_temp_prop,
  digits = 2,
  align = "c",
  caption = "Proportion of Species by Water Requirement within Each Temperature Class"
)
```

Table 1: Proportion of Species by Water Requirement within Each Temperature Class

Temperature_Class	Water	Count	Proportion
Half Hardy	Medium	11	0.92
Half Hardy	Very High	1	0.08
Hardy	High	10	0.15
Hardy	Low	13	0.19
Hardy	Medium	42	0.63
Hardy	Very Low	2	0.03
Tender	High	12	0.32
Tender	Low	1	0.03
Tender	Medium	23	0.62
Tender	Very High	1	0.03
Very Hard	Medium	1	1.00
Very Hardy	High	1	0.05
Very Hardy	Low	4	0.19
Very Hardy	Medium	16	0.76
Very Tender	High	2	1.00

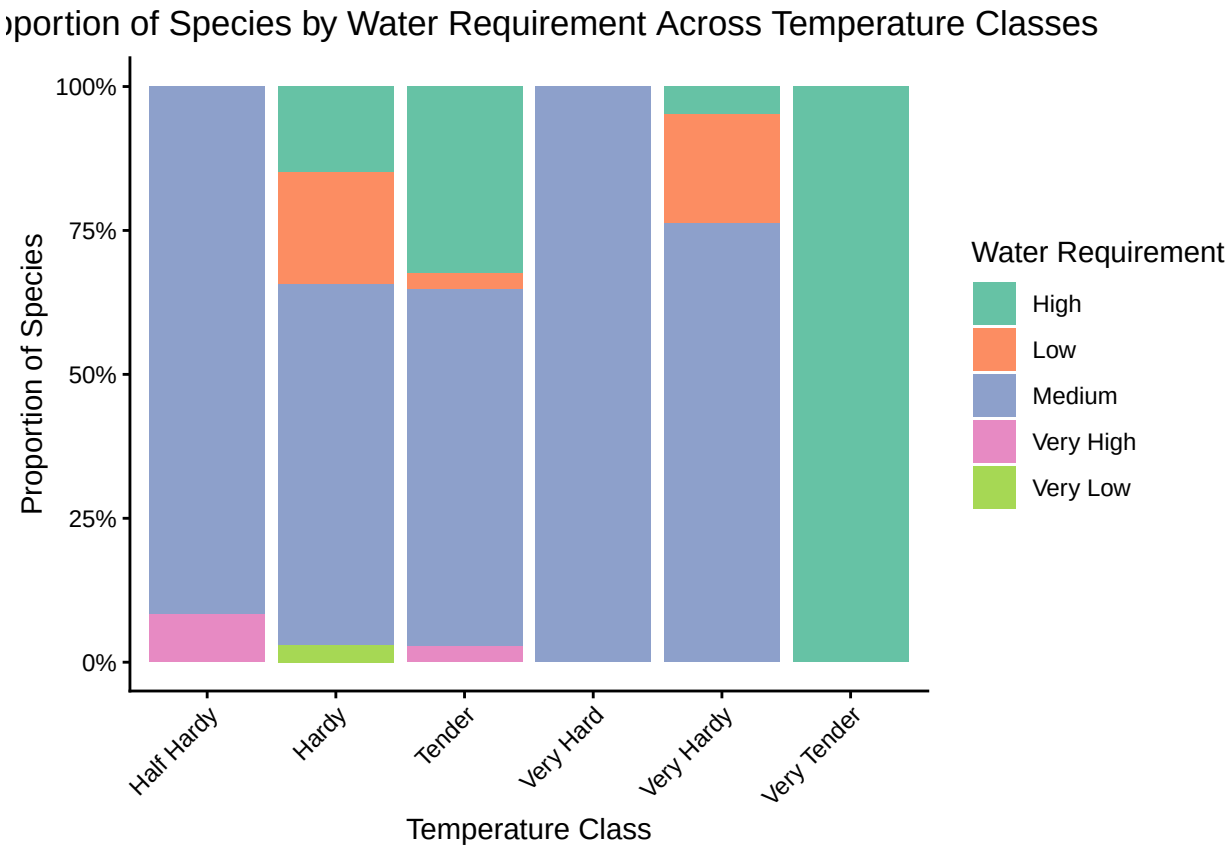
Most species with different temperature tolerances show distinct water requirements. Medium water requirement dominates across several temperature classes, while High, Low, Very High, and Very Low categories occur more selectively. These patterns indicate that water requirements vary across temperature tolerance classifications, suggesting a potential association between the two variables.

5 COMPARATIVE VISUALIZATIONS

```r
library(ggplot2)
library(dplyr)
library(scales)
library(RColorBrewer)

# Keep only observed categories
plot_data <- data_edible_plants_clean %>%
  group_by(temperature_class, water) %>%
  summarise(Count = n(), .groups = "drop") %>%
  group_by(temperature_class) %>%
  mutate(Proportion = Count / sum(Count)) %>%
  ungroup()

# Stacked bar plot without extra factor levels
ggplot(plot_data, aes(x = temperature_class, y = Proportion, fill = water)) +
  geom_bar(stat = "identity") +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    x = "Temperature Class",
    y = "Proportion of Species",
    fill = "Water Requirement",
    title = "Proportion of Species by Water Requirement Across Temperature Classes"
  ) +
  theme_classic() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
  )
```



The stacked bar plot shows how species' water requirements vary across temperature tolerance classes. Medium water requirement is the most common across most temperature classes, while High, Low, Very High, and Very Low categories appear less frequently and are often specific to certain classes. These patterns indicate that water requirements differ by temperature tolerance, supporting a potential association between the two variables.

6 AI USE DISCLOSURE STATEMENT

I used ChatGPT to help format the plots for a better, color blind friendly data visualisation.